

A pixel-trained world model contains a conserved quantity, and enforcing it makes the model predict better

Abstract

A world model that predicts video well may have learned something about the underlying physics, or it may simply be good at interpolating frames. Standard probes cannot tell the difference: even randomly initialised DreamerV3 models can contain latent directions that correlate strongly with a pendulum’s energy. We instead ask whether the model has learned a quantity that its *own dynamics* treat as conserved. In a frozen DreamerV3 trained only on 64×64 pendulum videos, with no physical supervision, we recover a scalar that correlates with true energy at 0.973 and remains nearly constant along encoded trajectories. When the model rolls forward without new observations, however, this quantity begins to drift.

We use three controls to test whether the recovered quantity reflects physics. First, six untrained models span energy correlations from 0.170 to 0.908, showing that high correlation alone is weak evidence; what training produces is reproducibility, with three trained models agreeing within 0.008. Second, when we apply the same procedure to models trained on a damped pendulum, where no conserved energy should exist, it correctly refuses to find one: four separate statistics distinguish conservative and dissipative models with no seed overlap. Third, the recovered quantity is causally useful. Projecting imagined states back toward its level set reduces rollout error on every trained seed, whereas sixty matched random polynomial constraints increase error by 53.9% at the median. The recovered law outperforms every random constraint on two of three models, though not on the third.

The internal representation is also less coordinate-specific than it first appears. Basis-independent properties of the recovered law agree closely across independently trained models, while selectors based on individual latent coordinates can even change sign. The same physical structure therefore appears in different internal charts. We also report two negative results. Requiring the recovered quantity both to stay conserved and to generate the model’s flow improves reproducibility but not accuracy, suggesting that conservation does most of the work here. And a frequency-weighted error-accumulation mechanism that we previously established in a much smaller recurrent model is indistinguishable from an unweighted version in DreamerV3.

Overall, a pixel-trained world model can learn a reproducible conserved quantity that its own dynamics respect, violate during free imagination, and can be causally corrected against, while also showing why correlation alone cannot establish that a model has learned physics.

1 Introduction

A world model that predicts video well might have learned physics, or it might have learned to interpolate video. Telling those apart matters if we want to plan with such a model, trust it out of distribution, or design a better one. The usual evidence is a probe: train a small readout from the hidden state to a physical variable, and report its accuracy. That evidence is weak, and its weakness has a clean demonstration in our own setting. Six randomly initialised DreamerV3 models, which have learned nothing at all, contain a direction correlating with the pendulum’s energy at up to 0.908.

The gap is that decodability says nothing about the model’s dynamics. A quantity can be readable from the hidden state while the model’s own transition map destroys it every step. What we want to know is whether the model’s dynamics *preserve* the quantity, and whether that preservation does any work.

We look for a **flow-conserved quantity**: a scalar C of the latent that stays constant under the model’s own one-step transition, found without ever showing the model a physical label. Three properties make it

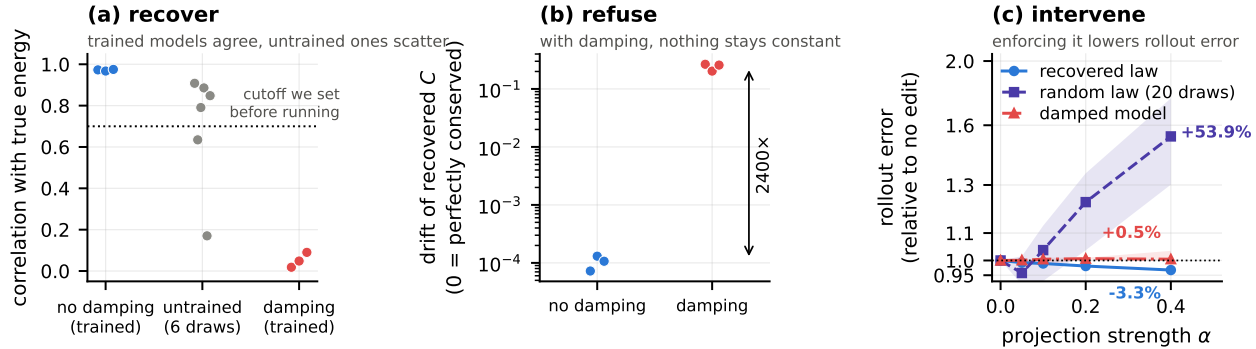


Figure 1: Each point is one seed. **(a)** The three trained models all land between 0.967 and 0.975, while six untrained ones scatter from 0.17 to 0.91 and four of them clear the cutoff we had fixed in advance. A high score alone therefore proves nothing; the agreement does. **(b)** The recovered quantity stays put on models trained without damping and drifts 2400 $\times$ more on models trained with it; these are the conservative and dissipative arms of Section 4. **(c)** Enforcing it lowers rollout error; enforcing a random polynomial of the same degree raises it by half, and enforcing it on a damped model does nothing.

checkable. It should be reproducible across seeds. It should vanish when the physics does. And enforcing it should improve the model’s predictions rather than damage them.

We test all three on a reference DreamerV3 [Hafner et al., 2023] trained on 64×64 video of a gymnasium pendulum, with no actor, no critic, and no physical supervision. After training we freeze the model and only read from it.

Our results are the following.

1. **Recovery is reproducible, and the level is not the evidence.** At a pre-registered extraction dimension, three unseen seeds recover a quantity correlating with true energy at 0.973, 0.967, and 0.975, with a within-trajectory drift of 10^{-4} . Six untrained models span 0.170 to 0.908. The trained seeds span 0.008 and the untrained draws span 0.738, so training buys consistency rather than the presence of the information.
2. **The method returns nothing when there is nothing to find.** On a matched dissipative model, chosen by a damping rule we fixed before training anything, four statistics separate with no overlap between seeds, including a 2400 $\times$ difference in drift.
3. **The recovered quantity is causally load-bearing, on two models of three.** Projecting the latent onto its level set during imagination lowers rollout error by 2.9–3.5% on every seed, while the same projection with a random polynomial of matched complexity raises it by 53.9% at the median of sixty draws. Against each model’s own twenty-draw null the recovered law is better than every random law on two seeds and at the 15th percentile on the third, where 7 of 20 random laws also improve the rollout. We report that split rather than the median.
4. **The correction acts on directions the latent does not make prominent, and it reproduces functionally without reproducing coordinate-wise.** The projection pushes hardest where a displacement most damages the model’s own rollout (+0.29 to +0.69 across seeds), and at a 40-step horizon those directions anti-correlate with variance (−0.36, −0.26, −0.41), an effect that decays with horizon. Across seeds the basis-free quantities agree closely (recovery to 0.008, edit slope to 0.016, gradient curvature to 0.046) while a selector defined on ranked coordinates changes sign. Identically trained models implement the same physical structure in different internal charts.
5. **Two things do not carry over, and we report them.** The flow-generation condition that motivates our selection rule turns out to buy reproducibility rather than accuracy: an ablation holding everything else fixed shows conservation alone recovers the energy nearly as well and edits just as effectively, but varies 4.4 $\times$ more across seeds. And a mechanism we established on a much smaller recurrent model

(Section 2), which predicts trajectory error from the accumulated invariant error weighted by the system’s frequency sensitivity, does not separate from its unweighted form at this sample size.

We wrote every criterion above down before the run that tested it, including the ones that failed. Section 7 reports a case where we refuted our own central claim on a bad statistic and had to withdraw the refutation.

2 Setup

We analyse a DreamerV3 world model [Hafner et al., 2023] trained on video of a pendulum. The model is the reference PyTorch implementation at its published architecture: 13.5M parameters, categorical 32×32 stochastic latents, a convolutional encoder and decoder, Kullback-Leibler (KL) balancing, and unimix. We train the world model alone, with no actor and no critic, because we never need a policy.

Relation to our earlier study. We developed this extraction procedure on a much smaller system, and we describe it here in full because there is no separate paper to cite. A gated recurrent unit (GRU) of about 13k parameters was trained to predict the next position of a simulated oscillator from position alone, so it read the state directly instead of inferring it from pixels. On that substrate the same search recovered a quantity correlating with true energy at 0.972 or better, scored close to zero on damped controls, and lowered rollout error by 68% when enforced. That study is also where the three controls in this paper come from: an untrained network, a dissipative system, and a random law of matched complexity.

Two of its findings are retested in Section 7 and neither survives here. We treat everything from it as a hypothesis to retest rather than as support, for two reasons. It is our own unpublished work, so a reader cannot check it. And the two substrates do not share an error metric: the GRU’s rollout error lives in the oscillator’s two-dimensional state space, while ours is pixel error against held-out video. Percentages from the two are not interchangeable, and Section 5 says so where it quotes one. Appendix A describes that study in full, including the results that did not survive it.

Code and run logs. Every experiment in this paper, the run logs its figures are drawn from, and the source of the paper itself are at <https://github.com/Zarand3r/latent-noether-paper1>. The figures regenerate from the committed logs with no GPU, so each number can be checked against the run that produced it. Model checkpoints and rendered video are too large to commit; the repository pins the upstream commit of the DreamerV3 implementation our adapter is built against and gives the commands to rebuild both.

Data. We render the video ourselves from gymnasium’s `Pendulum-v1` [Towers et al., 2024]. Each step produces a 500×500 RGB frame, which we crop to 448×448 and block-average to 64×64 . We write initial conditions directly into the environment state, sampling θ uniformly on $[-1.8, 1.8]$ and $\dot{\theta}$ on $[-2.2, 2.2]$, and we reject any trajectory that reaches the simulator’s $|\dot{\theta}| = 8$ speed clip, since that clip is a hard nonlinearity that would destroy conservation. Actions are always zero, so the pendulum evolves freely.

Velocity never appears in a single frame. There is no motion blur, no trail, and no torque arrow, so the model has to infer θ from the frame sequence on its own. That is what makes the latent worth examining.

Training. We train on a fixed dataset of 256 trajectories of 120 frames, using Adam at 10^{-4} , batch 16, sequence length 64, for a wall-clock budget of half an hour. That budget buys about 6,500 gradient steps, a small fraction of a standard DreamerV3 run. We set free bits to 0 rather than the reference default of 1.0; the measured KL settles near 1.8 nats, far from collapse, so the floor does no work here. Three seeds train independently and all three pass acceptance checks we registered beforehand: KL above 1 nat, one-step decoding at least $4\times$ better than predicting the dataset mean, and rollouts that stay finite. A model that failed to train would tell us nothing about the analysis, so we discard it rather than report it.

This regime is offline, world-model-only, zero-action, and short. Our claims are about a DreamerV3 world model fitted to pendulum video, and not about DreamerV3 as a reinforcement learning agent.

Extraction. After training we freeze the model. Everything that follows reads the hidden state and never updates a weight. We encode the held-out videos, take the deterministic recurrent state h , and project it onto its top 12 principal directions, dropping any direction the data does not occupy. Inside that subspace we take the flow F to be the model’s own one-step transition displacement, $T(h) - h$, rather than a vector field we fit ourselves. Fitting a field and then analysing it would interpret our own regression instead of the model.

We then search for a degree-4 polynomial C that is constant along the model’s trajectories. The search is a generalized eigenproblem: the invariance ratio, within-trajectory variance over total variance, is a Rayleigh quotient, so one eigendecomposition returns the whole ranked family. We fit C jointly with an antisymmetric B satisfying $F \approx B\nabla C$, which is the condition that C generates the flow rather than merely staying constant along it [Arnold, 1989]. Section 7 reports how much that second condition contributes, which is less than we expected.

What we compare against. We score C against the textbook energy $E = \frac{1}{6}\dot{\theta}^2 + 5\cos\theta$. Gymnasium integrates semi-implicitly, so it conserves a shadow Hamiltonian $\tilde{H} = H + O(\Delta t)$ rather than E exactly [Hairer et al., 2006]. We measure about 12% relative oscillation in E from that mismatch, with no secular drift. Ground truth enters only when we score a result, never inside the search.

3 The model contains a conserved quantity, and the level alone does not show it

At the extraction dimension we registered in advance, the recovered C correlates with true energy at $|\rho|_E = 0.973, 0.967$, and 0.975 across three seeds the choice never saw. The recovered quantity stays constant along the trajectories the model encodes from held-out video: its within-trajectory variance is 1.1×10^{-4} of its total variance, so it is nearly flat along a trajectory and varies between trajectories, which is what a conserved quantity does. Two cautions about that number. We measure it where an observation corrects each step; under the model’s own imagination the same C drifts far more, which is what Section 5 exploits. And it is a different statistic from the *held-out invariance ratio* of Table 1, which scores the invariance-only candidate rather than the jointly fitted C ; the two are not comparable and we report both.

That number on its own is close to meaningless, and the reason is the same one that makes probing weak evidence [Hewitt and Liang, 2019, Belinkov, 2022]. Our search returns functions that stay constant within a trajectory and vary between trajectories. On a one-degree-of-freedom conservative pendulum, energy is essentially the only such function, so any faithful encoding of $(\theta, \dot{\theta})$ will yield something correlated with it. A randomly initialised network encodes the frame history perfectly well.

An untrained model clears the gate we registered. We ran the identical pipeline on six randomly initialised DreamerV3 models. They score $|\rho|_E$ from 0.170 to 0.908, and four of the six clear the 0.7 threshold we had registered as the success criterion (Figure 1a). The threshold measured nothing about learning. We had carried it over from an earlier architecture without checking it against this one’s null.

The honest comparison is a difference, and it separates cleanly on two statistics at once. Every trained seed scores above every untrained draw (0.967 against 0.908), and every trained seed has a lower pairing residual than every untrained draw (0.865 against 0.912). Neither range overlaps.

Consistency separates trained from untrained more sharply than level does. The three trained seeds span 0.008 in $|\rho|_E$. The six untrained draws span 0.738, which is $92\times$ wider. A random network sometimes lands near 0.9 and sometimes at 0.17, depending on its initialisation. The trained models land at 0.97 every time.

Takeaway. Training does not create energy information in the latent, since an untrained network often has it. Training makes the recovered structure reproducible across seeds and conserved under the model’s own transition. That is the claim the evidence supports, and it is narrower than the one the raw correlation suggests.

Table 1: Handing the pipeline a dissipative model returns nothing conserved. Ranges are the three seeds per arm; no seed in one arm overlaps any seed in the other on any row.

	conservative	dissipative	median ratio
$ \rho _E$	0.967–0.975	0.018–0.090	20×
drift of C	$0.7\text{--}1.3 \times 10^{-4}$	0.203–0.266	2414×
held-out invariance ratio	0.0027–0.0121	0.983–0.994	282×
pairing residual	0.829–0.865	0.934–0.957	n/a

4 The method returns nothing when the physics is absent

A search that always returns a conserved quantity would explain the previous section without any physics. We test that directly by running the identical pipeline on a world model trained on a *dissipative* pendulum, where no non-trivial conserved function of the state exists: every trajectory spirals into the same fixed point, so any continuous function constant along trajectories is constant on the whole basin.

Choosing the damping without looking at the outcome. Strong damping collapses the trajectories into a small neighbourhood of the fixed point, which makes the latent rank-deficient and manufactures directions that are trivially constant. A refusal produced that way would say nothing about the method. We therefore fixed a rule before training anything: take the largest damping ratio whose late-window spread across trajectories stays above 25% of the conservative reference, measured on simulator states alone with no model involved. The natural choice $\zeta = 0.15$ fails that rule outright, leaving 0% of the reference spread. The rule selects $\zeta = 0.03$, which removes 17% of the energy per period and leaves 34% of the spread. All three dissipative models pass the same training acceptance checks as the conservative ones.

Four statistics separate with no overlap between seeds. Table 1 reports them. The recovered C drifts 2400× more on the dissipative models, and its held-out invariance ratio is 0.987, meaning the model conserves nothing. That is the right answer.

Takeaway. The pipeline discriminates conservative from dissipative dynamics under matched architecture, training budget, observation modality, and extraction protocol. The correlation in Section 3 is therefore not an artefact of a search that always returns something.

5 Enforcing the recovered law improves the model’s own predictions

If C is the quantity the model’s dynamics actually respect, then holding it fixed during imagination should remove drift that is pure error. If C is something we fitted, holding it fixed should damage the rollout. We test that by projecting the latent back onto the level set of C at every imagination step,

$$z \leftarrow z - \alpha (C(z) - C_0) \frac{\nabla C(z)}{\|\nabla C(z)\|^2}, \quad (1)$$

and mapping the correction back so the part of h outside the extracted subspace stays untouched. We retrain nothing. The model stays frozen and the edit happens at decoding time.

We score the slope over a fixed grid, not the best α . We sweep α over (0, 0.05, 0.1, 0.2, 0.4), identical for every arm, and report the whole curve. Taking the minimum over a grid once told us an arm had improved by 5.9% when its curve rose monotonically to twice its baseline error. The direction of the curve is the result; the minimum is a selection over five chances.

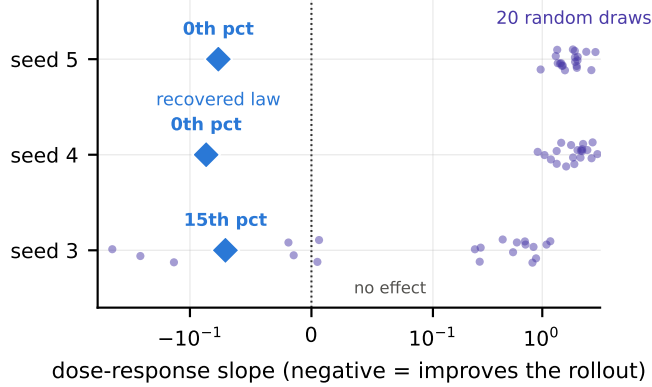


Figure 2: Each model against its own null. Twenty random degree-4 polynomials, matched in norm and identical across models, give the twenty small marks per row; the diamond is that model’s own recovered law. On seeds 4 and 5 no random law comes close. On seed 3 seven of them improve the rollout and the recovered law ranks 15th of 20, which is the limit of the specificity claim.

Three arms, and the random one is a distribution. The first enforces each conservative model’s own recovered C . The second enforces a random polynomial over the same degree-4 basis, matched in norm, on the same models. The third enforces each dissipative model’s own recovered C . The random arm is the one that can kill the result: if projection improves rollouts whatever we enforce, then the edit shrinks the rollout toward a manifold and has nothing to do with physics. That arm is therefore drawn twenty times per model rather than once, and scored as a null distribution with the recovered law’s percentile inside it. The same twenty polynomials run on every model, so a difference between models is the model and not the draw.

Figure 1c shows the outcome. The recovered law reduces rollout error by 2.9%, 3.3% and 3.5% at the strongest projection, on all three seeds. Across the sixty random laws the median is 53.9% *worse*, and only 7 of 60 improve the rollout at all. The same edit on the dissipative models does nothing, moving error by 0.5%, which is what should happen when the level set we enforce is not a law.

The arms differ more in variance than in median. The recovered law’s per-seed slopes span 0.016. The random draws’ slopes span 3.71, which is 234 $\times$ wider. Enforcing the recovered law does the same small good thing on every seed. Enforcing a random law does something different and usually harmful each time.

The specificity is strong on two seeds and weak on the third. Per model, the recovered law’s slope sits at the 0th percentile of that model’s own twenty draws on two seeds, better than every random law, and at the 15th percentile on the third (Figure 2). On that third model 7 of 20 random laws improve the rollout, and the best of them reaches -21.9% , seven times the improvement the recovered law achieves there. So the claim the evidence supports is that the effect is specific to the recovered law on two models of three, and that on the third a projection onto almost any smooth level set helps more. We do not know what distinguishes that model.

The scoring rule is why we can say that. The same seed’s best random law is not monotone: its error falls to $\alpha = 0.2$ and then rises. A single- α comparison would have reported a random law beating the recovered one; the fixed-grid slope rule reports the curve instead. This is the second time that rule changed a conclusion for us. It is also why the twenty-draw null was worth its cost: the earlier version of this experiment drew one polynomial and reused it on all three models, which made a single lucky or unlucky draw look like a three-seed result.

Takeaway. The recovered quantity is causally load-bearing for the model’s own rollouts, and the effect is specific to it. The magnitude is small. The GRU study measured 68% for what is procedurally the same edit, but the two numbers are not comparable: it scored error in the oscillator’s state space, while we score pixel error against held-out video, where most of each frame is static background that reconstructs correctly

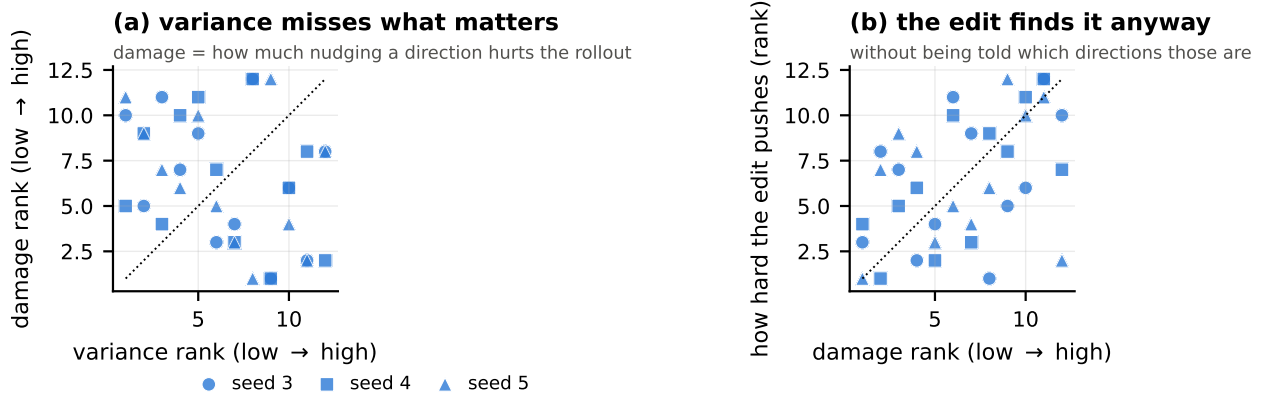


Figure 3: Each point is one of the twelve extracted directions, ranked within its own model; the dotted line marks perfect agreement. **(a)** The directions the latent gives the most variance are not the ones a nudge hurts most, on any seed (rank correlation -0.36 , -0.26 , -0.41 for seeds 3, 4, 5). **(b)** The projection of Section 5 nonetheless pushes hardest on exactly those high-damage directions ($+0.38$, $+0.69$, $+0.29$), though the correlation is far from perfect, so the correction is only partly targeted.

whatever the latent does. Part of the gap is therefore definitional. What remains could come from the architecture, the system, or the extraction dimension, and we have not separated those.

6 What the correction acts on

Section 5 shows the projection helps and that a random law of the same complexity hurts, which says the effect is specific without saying what it is specific *to*. This section measures that. For each direction u in the extracted subspace we record its statistical prominence and its causal leverage,

$$V(u) = \text{Var}(u^\top z), \quad D_H(u) = \mathbb{E}[L_H(z + \epsilon u) - L_H(z)], \quad (2)$$

where L_H is the model’s own H -step rollout error. V is what principal-component analysis reports. D_H is what the model’s dynamics say the direction is worth, and it needs no simulator.

Errors compound rather than decay. A displacement injected into the latent grows as the model rolls forward. After 40 steps it retains $6.3\times$ its initial size at $\epsilon = 0.05|z|$, $2.8\times$ at $\epsilon = 0.25|z|$ and $1.9\times$ at $\epsilon = 0.5|z|$, as medians over the three seeds; larger displacements amplify less, as a saturating nonlinearity does. A contracting transition would make small latent errors self-correcting and this whole section moot. This one amplifies them, which is why a bounded invariant error turns into an unbounded trajectory error.

Calibrating the displacement. Our first attempt used $\epsilon = 0.05|z|$ and produced damage near -0.5% of baseline with the sign varying between directions, which read as “variance predicts consequence at $+0.40$ ”. The rollout is exactly deterministic, so a repeated unperturbed rollout differs by zero and those values were not sampling noise: ϵ was too small for the response to leave the second-order regime. Damage against displacement for the leading direction runs -0.5% , $+3.1\%$, $+18.1\%$, $+46.4\%$, $+128.8\%$ at $\epsilon/|z|$ of 0.05, 0.10, 0.25, 0.50, 1.00. We fix $\epsilon = 0.25|z|$, the smallest displacement whose damage exceeds 10% of baseline while the response still grows smoothly.

The edit acts on the directions that matter. How hard the projection pushes on each direction correlates with D_H at $+0.385$, $+0.692$ and $+0.287$ across seeds, positive on all three (Figure 3b). The correction spends its effort where displacement does the most damage. The correlation sits well short of 1, so the correction is partly targeted and partly spread across the latent.

Variance misses those directions, at short horizons. At the same 40-step horizon, prominence and damage anti-correlate: -0.364 , -0.259 , -0.406 , negative on all three seeds, with damage spanning $13.5\times$ to $118.8\times$ between the most and least consequential direction (Figure 3a).

We checked whether that survives a change of horizon before drawing anything from it, and it does not survive far. Rank correlation of D_H between disjoint halves of the evaluation trajectories is 0.95, so the per-direction measurement itself replicates. The disagreement with variance does not:

horizon	20	50	100
median $\rho(V, D_H)$	-0.252	-0.196	-0.028
negative in	9/9 settings	5/9	5/9

At horizon 20 the anti-correlation holds for every seed and every displacement we tried. By horizon 100 it is gone, and one seed reaches $+0.727$. The explanation is simple: once a rollout has diverged from the truth, whichever directions carry the most amplitude dominate the total error, so variance and damage measure the same thing. The mismatch lives in the regime where the model still tracks the system.

What we have not established about it. Everything in this section measures the conservative models alone, so we cannot say whether the mismatch is a property of a model that learned a conserved quantity or of autoregressive prediction on smooth data generally. Separating those needs the same measurement on a matched model with no conserved quantity. We have that model (the dissipative arm of Section 4) and report the comparison as future work rather than folding an unreviewed result into this claim.

The correction reproduces functionally, but not coordinate-wise. Three models trained identically except for their seed agree closely on everything we can state without reference to a basis, and disagree on everything we cannot. Recovery agrees to 0.008, the edit’s dose-response slope to 0.016, and the curvature of the gradient field, measured as the top-3 eigenvalue mass of $M_C = \mathbb{E}[gg^\top / \|g\|^2]$, to 0.046 (0.448, 0.441, 0.487).

Now select a protected subspace two ways and score each by the same dose-response slope. Ranking the latent *coordinates* by ∇C and keeping the top three gives -0.075 , $+0.016$, -0.089 : it beats a variance ranking on two seeds of three, loses badly on the third, and changes sign, so it fails a margin we had registered on all three. Replacing the coordinate mask with an orthogonal projector onto the top eigenspace of M_C beats the variance selector on all three seeds. The coordinate-dependent selector varies by 0.105 across seeds where the basis-free quantities vary by 0.008 to 0.047; those are different quantities on different scales, so the comparison is suggestive rather than decisive, but the sign change is not a matter of scale.

We take that to mean the models implement the same physical structure in different internal charts. One limit on the subspace result, stated because it is easy to overread: on two seeds the gradient subspace also sits below the 5th percentile of a hundred random rank-3 subspaces, but on the third it sits at the 41st. It is better than variance on every seed and not yet shown to be better than an arbitrary subspace of the same rank.

That M_C has less than half its mass in three of twelve directions is worth stating on its own. The correction is rank one at every state, so this spectrum measures how much ∇C *rotates* as the state moves. It rotates a lot, and by the same amount in every model. The law-bearing direction curves through the latent instead of resting on a fixed set of axes.

Takeaway. The projection that improves rollouts acts on directions whose displacement damages the model’s own predictions, and at short horizons those are not the directions the latent makes prominent. Read together with the paragraph above, the practical lesson is that the reproducible object here is a function of the state, not a coordinate: asking which latent units carry the physics gives a different answer in every model, while asking which direction the recovered law points in gives the same one. We report this as a caution for coordinate-level readings of a learned latent, and note the limit of our own evidence: comparing subspaces *across* models is not well defined in raw latent coordinates, since each model’s basis is arbitrary, so what we show is agreement of basis-free scalars rather than coincidence of the subspaces themselves. We state this as a measurement rather than a design principle: the effect is weak, it decays with horizon, and

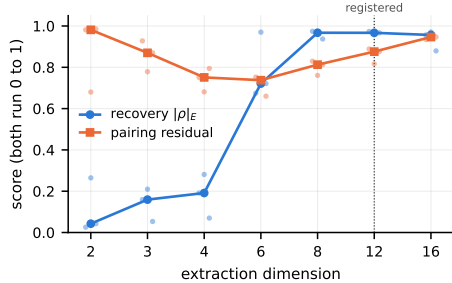


Figure 4: Recovery and the pairing residual move in opposite directions above dimension 6, so the residual cannot be read as a confidence score.

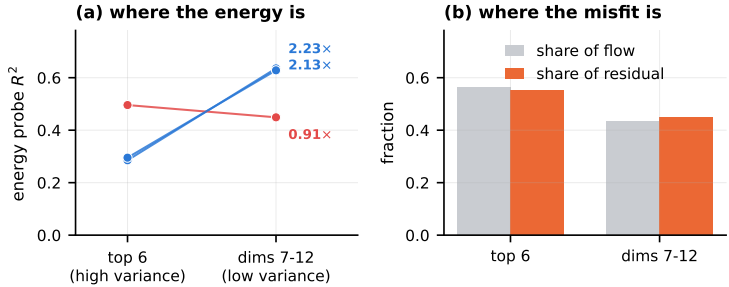


Figure 5: Medians over three seeds. The added low-variance directions carry twice the energy information of the top six (a) on two seeds of three and $0.91\times$ on the third, but on every seed take no more than their share of the misfit (b), which rules out the explanation we had registered for the anti-correlation on the left.

we have not shown that a model built to protect high-damage state predicts better than one built to protect high-variance state. That comparison is what would turn this into an architecture, and we have not run it.

7 What does not transfer

The flow-generation condition buys reproducibility, not accuracy. We select C by asking for two things: that it stay constant along the model’s trajectories, and that it generate the flow through $F \approx B\nabla C$. The second condition is the part that distinguishes this from a generic search for conserved quantities, and on this model it is close to inert. We compared the recovered C against 200 random degree-4 polynomials of matched complexity, scored on the same latent. The recovered C lands at the 17th to 66th percentile of that null on pairing residual, which is mid-pack. The dissipative models’ candidates land at the 100th percentile, worse than every random polynomial, so the statistic is not vacuous. On this model, conservation does the discriminating.

An ablation isolates what the term contributes. Holding the candidate family, data, dimension, degree and α grid fixed, we select C two ways: by conservation alone, taking the lowest-ratio eigenvector, and jointly with B as we do throughout. Conservation alone recovers the energy at 0.950 against the joint rule’s 0.973, and it edits marginally better (-0.087 against -0.077 , winning on two seeds of three). The one place the two differ is spread: the joint rule’s recovery varies by 0.008 across seeds and conservation alone varies by 0.035, $4.4\times$ wider. Flow-generation makes the extraction repeatable rather than more accurate, and it is not what makes the intervention work.

Figure 4 sharpens the problem. As the extraction dimension rises from 6 to 12, recovery improves from 0.721 to 0.967 while the pairing residual gets *worse*, from 0.738 to 0.876. Recovery is already at 0.967 by dimension 8, where the residual is better (0.813), so the dimension we registered is not where recovery arrives. It is past it. We registered the natural explanation for the anti-correlation, that the added directions carry the energy but resist a Hamiltonian fit, and it failed: on two seeds of three those directions carry more than twice the energy information of the first six ($2.23\times$ and $2.13\times$), but their share of the residual (44.9%) tracks their share of the flow (43.5%) almost exactly (Figure 5). On the third seed the energy relation inverts ($0.91\times$), which does not affect the residual result but does bound the claim below it. We cannot explain the anti-correlation, and we report the pairing residual as a diagnostic rather than as a confidence score.

Energy lives in the low-variance directions, on two seeds of three. Principal directions 7 through 12 carry $2.23\times$ and $2.13\times$ the energy information of the top six on two seeds, and $0.91\times$ on the third, where the relation inverts. We had seen the effect on a sharply anisotropic recurrent network, and it reproduces here on a near-isotropic stochastic model in two models of three. We state it per seed because a median would hide a sign change. It is the information-theoretic counterpart of the causal measurement in Section 6: variance misses the directions that carry the physics, and separately misses the directions whose displacement damages the rollout.

A mechanism from the GRU study does not replicate. The GRU study showed that phase error accumulates as the time integral of the invariant error weighted by frequency sensitivity, $\int \kappa(E_s) \Delta E(s) ds$ with $\kappa = d \ln \omega / dE$, beating a fitted power law on held-out data by $5\times$. Here the weaker part holds: accumulated invariant error predicts which rollouts drift most, with rank correlation rising to 0.503 as the horizon grows, and the integral predictor beats a fitted t^p (R^2 0.187 against 0.021). Shuffling destroys it. The frequency weighting specifically does not separate. Under a nested test with 400 bootstrap resamples, the weighted term contributes significantly beyond the unweighted integral in 1 of 9 seed-horizon combinations, and the unweighted beyond the weighted in 0 of 9. The two predictors are not distinguishable at $n = 512$.

We recorded this as a refutation once and were wrong. We had compared two standalone R^2 values differing by 0.019, never checked that gap against its sampling error, and let a three-seed median choose the direction while the weighted predictor actually led in 6 of 9 combinations. The verdict is unresolved.

Scope. One physical system, one degree of freedom, one architecture, and a training run far shorter than a standard DreamerV3 schedule. The pendulum straddles its separatrix, so 17% of trajectories rotate rather than oscillate and we restrict the mechanism analysis to the librating ones. We have not tested contacts, multiple objects, actions, or any system where the frequency varies smoothly enough to identify κ .

8 Related work

Probing shows a quantity is present, not that the model uses it. Linear and nonlinear probes read physical or linguistic variables out of hidden states [Alain and Bengio, 2017], and the literature has repeatedly shown that a probe’s accuracy confounds what the model encodes with what the probe can learn. Control tasks [Hewitt and Liang, 2019] and the surrounding critique [Belinkov, 2022] formalise that. Our untrained baseline in Section 3 is the same discipline applied to invariant search, and it matters more here than usual: four of six randomly initialised models cleared our registered threshold. Work on emergent world representations in sequence models [Li et al., 2023, Nanda et al., 2023] strengthens probing by intervening on the representation, which is the move we make in Section 5.

Equation discovery fits the data; we read the model. SINDy [Brunton et al., 2016] recovers governing equations from trajectories, and Hamiltonian and Lagrangian networks [Greydanus et al., 2019, Cranmer et al., 2020] build conservation into an architecture so that it holds by construction. Both target the physical system. We target a model that we never told about physics and trained only to predict pixels, and we ask what its transition map preserves. Koopman and dynamic-mode methods [Schmid, 2010] share our linear-algebraic machinery, and we use the standard version of it rather than proposing a new one.

The field evaluates world models on returns, not on physics. DreamerV3 [Hafner et al., 2023], its predecessors [Ha and Schmidhuber, 2018], and planning-based learned models [Schrittwieser et al., 2020] report task performance. That leaves open whether a model that predicts well has internalised any dynamical structure. The disentanglement literature [Locatello et al., 2019] showed how easily an appealing structural claim about representations survives without the right controls, which is the failure mode our dissipative arm guards against.

Positioning. The extraction machinery here is standard. The contribution is the pairing of a claim about a modern pixel-trained world model with the three controls that make it falsifiable: an untrained model, a dissipative model, and a random law.

9 Conclusion

A DreamerV3 trained only to predict pendulum video holds a quantity that its own dynamics conserve, that disappears when we remove the physics from the training data, and that improves its rollouts when we enforce it. The evidence is the separation from three controls, not the correlation itself, because an untrained model clears the correlation threshold we had registered.

The result suggests a design question we cannot answer yet. If the model carries a physical quantity and lets it drift, and if correcting that drift from outside improves prediction, then we might build a world model that preserves it during imagination instead. Whether that helps at a scale where the physics is not a single pendulum is the experiment we would run next.

A The GRU precursor study

The method in this paper was developed on a much smaller recurrent model before it met a pixel world model. That study is unpublished and has no separate reference, so we describe it here in enough detail to be judged. Its role is not to support the DreamerV3 result. It is where the phenomenon first appeared, where several appealing readings of it were killed, and where the narrower claim tested in the main text came from.

Setup. A single-layer GRU of 48–64 hidden units, about 13k parameters, trained on next-observation prediction for low-dimensional mechanical systems. The model saw *position only*, so any representation of velocity had to be reconstructed internally by the recurrent state. We studied two conservative systems, an anharmonic central-force system and a pair of coupled anharmonic oscillators, with matched dissipative variants trained as controls. Extraction never used a physical label; ground truth entered only when scoring a recovered quantity.

The search. Writing z for a low-dimensional representation of the hidden state and f for the latent flow induced by the trained transition, we look jointly for a scalar C and an antisymmetric field B with $f \approx B\nabla C$. Antisymmetry gives $\nabla C^\top B \nabla C = 0$, so any C satisfying the fit is approximately constant along the flow. We score the fit by the normalised residual $R = \|f - B\nabla C\|^2 / \|f\|^2$.

A.1 What it established

Conservation alone is not evidence. A randomly initialised GRU already contained a latent scalar correlating with true energy at $|\rho| \approx 0.998$. A high-dimensional random encoding of a low-dimensional trajectory retains enough information for a flexible readout to reconstruct energy. This is the result that forced every later claim to be a *difference* against a null rather than a level, and it is why the main text leads with the same control on DreamerV3.

Flow-based selection separated conservative from dissipative models. Once candidates were selected by the joint criterion, the recovered scalar correlated with true energy at $|\rho_E| \geq 0.972$ on conservative models, often approaching 0.999, and the residual separated the two arms strongly ($p = 5.4 \times 10^{-6}$, $d = 3.10$ on a representative comparison).

Enforcing the recovered quantity improved rollouts. Correcting the hidden state during free rollout so that C stayed near its initial value reduced trajectory error by up to 68% and true-energy drift by 69%, and *harmed* dissipative models. This is the first version of the recover, refuse, intervene chain that the main text retests.

Long-horizon error is mostly phase. About 77% of rollout error lay along the true physical flow and only 14% changed the conserved quantity. The model stayed near the right orbit and moved along it at the wrong rate.

The frequency–action mechanism. In a nonlinear oscillator the frequency depends on the conserved quantity, so a small energy error produces a persistent frequency error that accumulates as phase. With $\kappa(E) = d \ln \omega / dE$ and $X(t) = \int_0^t \kappa(E_s) \Delta E(s) ds$, this predictor beat generic functions of time on held-out data.

predictor	central force	coupled oscillators
$a + bt$	0.124	-0.028
$a + bt^p$ (best p)	0.131	-0.024
$a + b \int \kappa \Delta E ds$	0.658	0.358
with an oscillatory correction	0.703	—
with an angular-momentum term	0.657	—
fitted coefficient b	0.91	0.67

The coefficient sits near the natural scale the mechanism predicts, and it was never fitted for that purpose: κ came from the simulator. Adding an oscillatory correction raises the central-force fit to 0.703, while adding an angular-momentum term moves it to 0.657, slightly worse, and the coupled system has no rotational symmetry at all. The scalar-energy account is therefore already complete for the errors these models actually make. A one-time intervention on the invariant, after which the model ran free, changed later phase drift with the predicted sign in 94% of cases and a held-out R^2 of 0.887.

A.2 What it killed, and why that shapes the main text

∇C is not the physical energy gradient. Matching perturbation directions so each produced the same instantaneous change in true energy did *not* equalise their downstream effects. Displacing along ∇C moved later phase drift most per unit of energy change, a random direction less, and a direction tangent to C least, on every seed. The size of the gap depends on how the coefficient is estimated, so we report the ordering rather than the digits; what matters is that instantaneous ΔE is not a sufficient mediator. Separately, ∇C moved true energy only $1.08\times$ more per unit displacement than a random direction. A model can hold the right scalar while its surrounding latent geometry is wrong.

A symmetry the model gets wrong. For a rotationally symmetric central force the radial frequency must be even in angular momentum, $\omega_r(E, L) = \omega_r(E, -L)$. Measured at fixed energy, the simulator obeys this: $|L|$ alone predicts its radial frequency at $R^2 = 0.89$, and signed L adds nothing ($R^2 = 0.00$). The learned GRU does not. Signed L predicts its frequency at $R^2 = 0.30$, a dependence the physics forbids exactly, and it is the larger of the two effects the model shows. Recovering an energy-like scalar is therefore not the same as recovering the law.

Structural readings that failed. A low-dimensional part of the hidden state carried most of the information reaching decoded futures, which suggested a quotient interpretation; the corresponding divergence statistic then separated conservative from dissipative systems at only $\text{AUC} = 0.68$, so we keep the observation as description rather than explanation. A subspace chosen by transition-closure criteria did not beat plain PCA at matched dimension. Extraction was also sensitive to representation: whitening the latent could destroy recovery, and keeping only the highest-variance directions removed physical information, which is the result that reproduces most strongly in the main text.

A.3 What transferred

finding	GRU	DreamerV3 (this paper)
energy-like invariant recoverable	$ \rho_E \geq 0.972$	replicates, ≈ 0.97
untrained nets score high	up to 0.998	replicates, up to 0.908
conservative vs dissipative separate	$p = 5.4 \times 10^{-6}$	replicates, no seed overlap
enforcing it improves rollouts	up to 68%	direction only, 3.3%
residual predicts how physical C is	works	does not transfer
frequency weighting earns its place	R^2 0.658 vs 0.131	unresolved
∇C is the physical direction	false	remains unsupported
right scalar implies right geometry	false	motivates the chart reading

Two rows carry most of the weight. The intervention transfers in direction and specificity but not in magnitude, and Section 5 explains why the two percentages are not comparable. The flow-generation residual,

which was the study’s most distinctive instrument, does not transfer at all: on DreamerV3 it is mid-pack against random polynomials, and conservation does the work instead. The value of the precursor for this paper is that the DreamerV3 result did not come from one successful experiment. It came from a programme that first produced the phenomenon and then removed several tempting readings of it before the surviving claim was put to a standard pixel world model.

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